

CONTROL STRUCTURES**DAY 2****1. Sum Without '+' Operator****Problem:**

Given two integers a and b, find their sum **without using + operator**.

Input:

Two integers a b

Output:

Sum of a and b

2. Swap Using XOR**Problem:**

Swap two numbers using **bitwise XOR operator**.

Input:

Two integers a b

Output:

Swapped values

3. Count Digits**Problem:**

Given an integer n, count total number of digits.

Input:

One integer n

Output:

Number of digits

4. Reverse Any Number**Problem:**

Reverse a number of any length.

Input:

Integer n

Output:

Reversed number

5. Palindrome Number

Problem:

Check whether a number is palindrome.

Input:

Integer n

Output:

Yes or No

6. Armstrong Number (3-digit)

Problem:

Check whether a number is an Armstrong number.

Input:

Integer n

Output:

Yes or No

7. Strong Number

Problem:

A number is strong if sum of factorial of digits = number.

Input:

Integer n

Output:

Yes or No

8. Factorial of Number

Problem:

Find factorial of a number.

Input:

Integer n

Output:

Factorial of n

9. GCD of Two Numbers

Problem:

Find GCD (HCF) of two numbers.

Input:

Two integers a b

Output:

GCD

10. LCM of Two Numbers

Problem:

Find LCM of two numbers.

Input:

Two integers a b

Output:

LCM

11. Prime Number Check

Problem:

Check whether a number is prime.

Input:

Integer n

Output:

Prime or Not Prime

12. Sum of N Natural Numbers

Problem:

Find sum of first n natural numbers.

Input:

Integer n

Output:

Sum

13. Fibonacci Series (N terms)

Problem:

Print first n Fibonacci numbers.

Input:

Integer n

Output:

Series

14. Count Even and Odd Digits**Problem:**

Count even and odd digits in a number.

Input:

Integer n

Output:

Even count and odd count

15. Sum of Digits (General Case)**Problem:**

Find sum of digits of any number.

Input:

Integer n

Output:

Sum

16. Power Without Math.pow()**Problem:**

Calculate a^b without using built-in function.

Input:

Two integers a b

Output:

Result

17. Check Perfect Number

Problem:

A number is perfect if sum of its proper divisors equals the number.

Input:

Integer n

Output:

Yes or No

18. Binary to Decimal**Problem:**

Convert binary number to decimal.

Input:

Integer (binary format)

Output:

Decimal number

19. Decimal to Binary**Problem:**

Convert decimal number to binary.

Input:

Integer n

Output:

Binary representation

20. Find Largest Digit**Problem:**

Find the largest digit in a number.

Input:

Integer n

Output:

Largest digit

21. Print Natural Numbers**Problem:**

Print all natural numbers from 1 to 100.

Input:

No input

Output:

Numbers from 1 to 100 (space-separated)

22. Reverse Natural Numbers**Problem:**

Print numbers from 100 to 1.

Input:

No input

Output:

Numbers from 100 to 1

23. Print Alphabets**Problem:**

Print all lowercase alphabets from a to z.

Input:

No input

Output:

a b c ... z

24. Sum of Natural Numbers**Problem:**

Find sum of numbers from 1 to 100.

Input:

No input

Output:

Single integer (sum)

25. Sum of Even Numbers**Problem:**

Find sum of all even numbers from 1 to 100.

26. Sum of Odd Numbers

Problem:

Find sum of all odd numbers from 1 to 100.

27. Count Digits

Problem:

Given an integer n, count total digits.

Input:

Integer n

Output:

Digit count

28. First and Last Digit

Problem:

Find first and last digit of a number.

29. Sum of First and Last Digit

Problem:

Find sum of first and last digit.

30. Swap First and Last Digit

Problem:

Swap first and last digit of a number.
